Practical 03: Activity Lifecycle and Logcat

Source: Android Programming Practical(1).pdf | Source pages: 29–30

Aim

To develop and test an Android application demonstrating activity lifecycle and logcat.

Steps to Perform the Practical

1. Create a project for demonstrating the Activity Lifecycle.

2. Open MainActivity.java and override the lifecycle methods shown in the source practical.

3. Add Log.d() statements to each method so the lifecycle events appear in Logcat.

4. Run the app on an emulator or Android device.

5. Open Logcat in Android Studio and filter using the tag used in your code.

6. Observe the messages when the activity starts, resumes, pauses, stops and is destroyed. Rotate or leave the app if appropriate to trigger more events.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

C. Develop an Android application demonstrating complete Activity Lifecycle methods and analyze execution using Logcat.

Project Name: logcat

```
package com.example.logact;

import android.os.Bundle;
import android.util.Log;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {



    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    protected void onStart() {
        super.onStart();
        Log.d("on start invoked", "on started");

    }

    @Override
    protected void onPostResume() {
        super.onPostResume();
        Log.d("on resume invoked", "on resume");
    }

    @Override
    protected void onPause() {
        super.onPause();
        Log.d("on pause invoked", "on pause");
    }

    @Override
        protected void onStop() {
```

```
            super.onStop();
            Log.d("on stop invoked", "on stopped");
        }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d("on destroy invoked", "on destroy");
    }
}
```